alarm calls. The selfish gene theory has to come up with a convincing advantage of giving alarm calls which is big enough to counteract this danger.

In fact this is not very difficult. Bird alarm calls have been held up so many times as 'awkward' for the Darwinian theory that it has become a kind of sport to dream up explanations for them. As a result, we now have so many good explanations that it is hard to remember what all the fuss was about. Obviously, if there is a chance that the flock contains some close relatives, a gene for giving an alarm call can prosper in the gene pool because it has a good chance of being in the bodies of some of the individuals saved. This is true, even if the caller pays dearly for his altruism by attracting the predator's attention to himself.

If you are not satisfied with this kin selection idea, there are plenty of other theories to choose from. There are many ways in which the caller could gain selfish benefit from warning his fellows. Trivers reels off five good ideas, but I find the following two of my own rather more convincing.

The first I call the *cave* theory, from the Latin for 'beware', still used (pronounced 'kay-vee') by schoolboys to warn of approaching authority. This theory is suitable for camouflaged birds that crouch frozen in the undergrowth when danger threatens. Suppose a flock of such birds is feeding in a field. A hawk flies past in the distance. He has not yet seen the flock and he is not flying directly towards them, but there is a danger that his keen eyes will spot them at any moment and he will race into the attack. Suppose one member of the flock sees the hawk, but the rest have not yet done so. This one sharp-eyed individual could immediately freeze and crouch in the grass. But this would do him little good, because his companions are still walking around conspicuously and noisily. Any one of them could attract the hawk's attention and then the whole flock is in peril. From a purely selfish point of view the best policy for the individual who spots the hawk first is to hiss a quick warning to his companions, and so shut them up and reduce the chance that they will inadvertently summon the hawk into his own vicinity.

The other theory I want to mention may be called the 'never break ranks' theory. This one is suitable for species of birds that fly off when a predator approaches, perhaps up into a tree. Once again, imagine that one individual in a flock of feeding birds has spotted a predator. What is he to do? He could simply fly off himself, without warning his colleagues.